



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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SECOND REVISION EXAMINATION - 2025

[Time Allowed : 3.00 Hours]

BIOLOGY**PART - I (Bio-Botany) (Marks : 35)****[Max. Marks : 70]****SECTION - I****Note. (i) Answer All the questions.****8×1=8****(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.**

- "Gametes are never hybrid". This is a statement of
 - Law of Dominance
 - Law of independent assortment
 - Law of Segregation
 - Law of random fertilization
- Assertion (A) :** Gamma rays are generally use to induce mutation in wheat varieties.
Reason (R) : Because they carry lower energy to non - ionize electrons from atom.
 - A and R are correct
 - A is correct, R is wrong
 - A is wrong, R is correct
 - A and R are wrong
- An eminent Indian Embryologist is
 - S.R. Kashyap
 - P. Maheswari
 - M.S. Swaminathan
 - K.C Mehta
- In soil, Water available for plants is -----
 - Gravitational water
 - Chemically bound water
 - Capillary water
 - Hygroscopic water
- Match the following.

1. Totipotency	i) Reversion of mature cells into Meristem
2. Dedifferentiation	ii) Bio - chemical and structural changes of cells
3. Explant	iii) Properties of living cells develop into entire plant.
4. Differentiation	iv) Selected plant tissue transferred to culture medium

 - 1 - (ii), 2 - (i), 3 - (iv), 4 - (iii)
 - 1 - (iii), 2 - (i), 3 - (iv), 4 - (ii)
 - 1 - (iii), 2 - (i), 3 - (iv), 4 - (ii)
 - 1 - (i), 2 - (iii), 3 - (ii), 4 - (iv)
- Depletion of which gas in the atmosphere can lead to an increased incidence of skin cancer?
 - Ammonia
 - Methane
 - Nitrous oxide
 - Ozone
- The active principle trans-tetra hydro canabial is present in
 - Opium
 - Curcuma
 - Marijuana
 - Andrographis
- Pick out the odd pair.

a) Mass Selection	- Morphological characters
b) Pureline Selection	- Repeated self pollination
c) Clonal Selection	- Sexually propagated
d) Natural Selection	- Involves nature

SECTION - II**II. Answer any four questions.****4×2=8**

- Pyramid of energy is always uprights. Give reasons.
- What are the enzymes you can use to cut terminal end and internal phosphodiester bond of nucleotide sequence?
- Write the Botanical name and family name of Nilavembu. Write any one uses.
- What is Heterosis?
- Write short notes on Castor Aruna.
- Draw and Mark the parts of pBR322

SECTION - III**III. Answer any three questions. Question No. 19 is compulsory.****3×3=9**

- Write the differences between Habitat and Niche.
- List out the Functions of Tapetum.
- Mention the name of man - made cereal. How it is formed?
- Give the significance of Ploidy.
- Give an account on Cryopreservation.

SECTION - IV**IV. Answer all the following:****2×5=10**

- Discuss the steps involved in Microsporogenesis. (OR)
 - Write the difference between Continuous and Discontinuous Variation. (OR)
- Enumerate the anatomical adaptation of Xerophytes.
 - How will you prepare an organic pesticide for your home garden with the vegetables available from your Kitchen?



PART - II (BIO-ZOOLOGY) (Max. Marks: 35)

SECTION - I

Note: (i) Answer all the questions

(ii) Choose the most appropriate answer from the given four Alternative write the option code and the corresponding answer.

- SMOG is derived from
 - Smoke
 - Fog
 - Both A and B
 - Only A
- Which of the following forests is known as the "Lungs of the planet earth"?
 - Tundra Forest
 - Rain Forest of North East India
 - Taiga Forest
 - Amazon Rain Forest
- Match the following and Choose the correct combination from the options given below.

Column - I	Column - II
A. Mutualism	1) Lion and Deer
B. Commensalism	2) Round worm and man
C. Parasitism	3) Birds compete with squirrels for nuts
D. Competition	4) Sea Anemone on Hermit Crab
E. Predation	5) Barnacles attached to whales

 - A - 4 B - 5 C - 2 D - 3 E - 1
 - A - 3 B - 1 C - 4 D - 2 E - 5
 - A - 2 B - 3 C - 1 D - 5 E - 4
 - A - 5 B - 4 C - 2 D - 3 E - 1
- The first transgenic cow is
 - Dolly
 - Thum
 - Sameer
 - Rosie
- The process which the sperm undergoes before penetrating the ovum is
 - Spermiation
 - Cortical reaction
 - Sperminogenesis
 - Capacitation
- Co - dominant blood group is
 - A
 - AB
 - B
 - O
- Lac Operon model was proposed by
 - Watson and Crick
 - Jacob and Monod
 - Hershey and Chase
 - Korenberg and Khorana
- B cells that produce and release large amounts of Antibody are called
 - Memory cells
 - Basophils
 - Plasma cells
 - Killer cells

SECTION - II

II. Answer any four questions.

- Freshwater sponges follow a peculiar mode of Asexual Reproduction. Why?
- Differentiate between Tubectomy and Vasectomy?
- Why it is very difficult to prepare a vaccine for Common Cold?
- What are Sacred Groves?
- What is Haplo Diploidy?
- Write the central dogma of Protein Synthesis.

SECTION - III

III. Answer any 3 questions. (Question No. 19 is compulsory)

- Draw a neat labelled Structure of tRNA?
- Suggest a prophylactic measure for RH Factor incompatibility for a pregnant RH - mother?
- Differentiate between Homologous and Analogous Structures?
- "Ecoli is not suitable for Interferon production" Why?
- Write short notes on
 - Mayor Rohitansky Syndrome
 - Chronionic Villi Sampling

SECTION - IV

IV. Answer in Detail.

- List out the reasons for the richness of Biodiversity in the Tropical Countries? (OR)
 - If a person has AIDS, How will you diagnose it and Write about its preventive measure
- List down the effects and control of Noise Pollution. (OR)
 - Explain in detail the Aerobic and Anaerobic microbes involved in Bioremediation.